

MD SAYEED ABID

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AI/ML Software Engineer with experience building production-oriented Python, LLM, RAG, and AI workflow solutions using LangChain, Hugging Face, PyTorch, vector databases, Docker, and Linux. Experienced in designing reliable AI pipelines with semantic retrieval, prompt orchestration, structured-output validation, error handling, fallback mechanisms, and systematic evaluation of model failures. Strong background in debugging ML workflows, analyzing failure patterns, and translating technical findings into production-ready solutions in collaboration with engineering and business stakeholders.

Experience

ERA Environmental Software Solution, Montreal, Canada

Feb. 2024 – Aug. 2025 **MITACS Accelerate Research Intern**

- Built production-oriented Python and LLM workflows for enterprise document processing, integrating OCR, prompt orchestration, structured-output validation, error handling, and fallback mechanisms to improve reliability across heterogeneous inputs.
- Implemented RAG-based semantic retrieval using Hugging Face embeddings and FAISS/Qdrant, including metadata filtering and retrieval evaluation to connect incoming information with relevant enterprise knowledge.
- Investigated incorrect and low-confidence LLM/extraction outputs through systematic error analysis, refining prompts, validation rules, retrieval logic, and fallback components to improve workflow reliability.
- Developed automated Python workflows that converted unstructured artifacts into validated structured outputs for integration with internal enterprise systems.
- Collaborated with engineering and business stakeholders to investigate workflow failures, validate corrective changes and document operational procedures, known limitations and production behavior.

Concordia University, Montreal, Canada

Jan. 2023 – Aug. 2025 **Graduate Researcher**

- Designed, trained, and evaluated deep learning architectures in PyTorch for computer vision and document understanding, conducting controlled experiments across model architectures, hyperparameters, and large-scale datasets.
- Evaluated model reliability through systematic error analysis, benchmarking, and failure-pattern analysis, identifying low-confidence outputs and model failure modes and translating findings into corrective improvements.
- Managed reproducible and version-controlled experiments using Python, PyTorch, Git, TensorBoard, and Weights & Biases, tracking, metrics, checkpoints, and failure patterns.

City Bank PLC, Dhaka, Bangladesh

Jul. 2022 – Dec. 2022 **Assistant Officer**

- Analyzed confidential transaction and payment datasets to support fraud-monitoring workflows, helping identify suspicious activity, reduce false positives, and improve operational decision-making.
- Built Python/SQL ETL and feature-engineering pipelines for enterprise payment datasets, supporting reproducible ML training and downstream analytics.
- Translated model results into risk-based recommendations to support fraud monitoring and payment processing decisions.

Doer Services PLC, Dhaka, Bangladesh
May 2021 – Jul. 2022 **Software Engineer**

- Developed Python-based machine-learning solutions for client business problems, performing data preparation, model development, evaluation, and integration into analytical workflows.
- Translated client needs into technical solutions and created visual dashboards using Power BI prototypes to summarize insights for stakeholders.

Education

Concordia University, Montreal, Canada

Jan. 2023 – Aug. 2025 **MSc in Computer Science (Specialization: Machine Learning and Computer Vision)**

North South University, Dhaka, Bangladesh

Jan. 2017 – Apr. 2021 **BSc in Computer Science & Engineering (Specialization: Artificial Intelligence)**

Skills

- **AI & Agentic Systems:** Python, LangChain, LLM APIs, RAG, AI Agents, Prompt Engineering, Structured Outputs
- **Retrieval & Knowledge Systems:** Embeddings, Semantic Search, Vector Search, FAISS, Hugging Face
- **Applied ML & LLMs:** Predictive Modeling, Classification, Regression, Forecasting, Optimization, Recommendation Systems, Model Evaluation, Feature Engineering, PyTorch, scikit-learn, pandas, NumPy
- **Systems & Development:** Linux, Docker, Bash, REST APIs, Git, SQL
- **Cloud:** Azure OpenAI Service, Azure Databricks, GPU Clusters
- **Experimentation & Analytics:** Hypothesis Testing, Statistical Analysis, KPI Analysis, Error Analysis, Benchmarking, Weights and Biases.

Research Projects and Publications

- **Tramba:** Hybrid Vision-Mamba and Transformer architecture for table understanding (Ongoing)
- Generalization of Bangla Sign Language Recognition (IEEE Access Volume: 9)
- Local Climate Zone Mapping Using Clustering Algorithms (IGARSS 2022)

Projects

- **Document AI/LLM Pipeline:** Built a Python-based complex document parsing ML pipeline with OCR and LLM orchestration to reduce manual workload.
- **Tramba: Table Understanding:** Conducted large-scale benchmarking and evaluation on the PubTables-1M dataset using reproducible experimentation pipelines and performance analysis across detection and structure-recognition tasks.
- **AI Troubleshooting Agent:** Built a Python agent that analyzes Linux and Docker application logs, extracts failure signatures, retrieves related incidents and technical documentation using embeddings, and generates ranked root-cause hypotheses with recommended corrective actions.
- **ML/classification project:** Built supervised ML models using Python, scikit-learn, various predictive models such as CatBoost, Random Forest, LightGBM to predict business outcomes from structured datasets. Performed EDA, feature engineering, class imbalance handling, model evaluation, confusion-matrix analysis, and performance reporting.